

AROON KUMAR

Artificial Intelligence Engineer

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SUMMARY

I am an AI/ML Engineer with substantial hands-on experience developing production-ready intelligent systems leveraging LLMs, computer vision, and various machine learning methodologies. I have a proven record in creating effective RAG pipelines and multi-agent architectures while being proficient in relevant programming languages and frameworks. My proficiency in deploying models on edge devices adds to my contribution potential in innovative technology.

EXPERIENCE

AI/ML Solutions Executive Prudential Solutions

AI/ML Solutions Executive (Intern → Part-time)

08/2025 – 02/2026 Islamabad, Pakistan

Completion Letter: [View on Google Drive](#)

- Built and deployed voice chatbots, speech-to-speech AI systems, and LLM applications
- Led development of ITP ConvertX multi-city chatbot supporting 5 cities and 3 languages
- Designed RAG pipelines for document ingestion and dynamic knowledge retrieval
- Implemented multi-model AI architecture (GPT + Mistral fallback) for reliability
- Built Redis-powered API Gateway, reducing latency 50% for 200+ users
- Automated CI/CD pipelines, reducing manual work by 70%

Data Science Intern 10Pearls

Data Science Intern

12/2025 – 02/2026 Islamabad, Pakistan

Completion Certificate: [View on Google Drive](#)

- Developed end-to-end AQI forecasting system using ML models
- Built automated ML pipelines for feature engineering, training, inference
- Implemented CI/CD workflows for daily model retraining and deployment
- Achieved MAE <10% and R² >0.85 in predictions
- Built Interactive Streamlit dashboard with Explainable AI (SHAP, LIME)

PROJECTS

Fiber Wire Defect Inspection System

[PyTorch](#) · [OpenCV](#) · [VGG16](#) · [ONNX](#) · [Docker](#) · [Flask](#)

01/2026 – Present Islamabad, Pakistan

- Built explainable defect detection using VGG16 transfer learning; classified fiber wire images as Good/Anomaly with 94.7% accuracy
- Implemented heatmap-based defect localization generating bounding boxes without pixel-level labels
- Trained on MVTec AD Cable dataset with class weighting; fine-tuned last 3 conv layers with Cross-Entropy loss
- Exported model to ONNX and served via Dockerized Flask API for production deployment

AigilityX AI Platform

[RAG](#) · [STT](#) · [TTS](#) · [Agent Orchestration](#) · [Supabase](#) · [React](#)

08/2025 – 03/2026 Islamabad, Pakistan

<https://aigilityx.com/>

- Full-stack AI platform with modular services: RAG retrieval, speech-to-text, LLM reasoning, TTS and orchestration layers backed by Supabase
- Service-based architecture for extensible AI workflows and independent scaling

Air Quality Index Forecasting System

[Python](#) · [Scikit-learn](#) · [Streamlit](#) · [SHAP](#) · [LIME](#)

12/2025 – Present Islamabad, Pakistan

<https://air-quality-ml-mlops.streamlit.app/>

- ML-powered AQI forecasting predicting air quality for the next 3 days with automated pipelines
- Achieved MAE <10% and R² >0.85; deployed Streamlit dashboard with Explainable AI

KEY ACHIEVEMENTS



High Accuracy in AI Projects

Successfully developed complex AI models achieving high accuracy across various projects, demonstrating strong skills in AI/ML development and implementation.

SKILLS

Deep Learning Computer Vision Python

C/C++ PyTorch TensorFlow Keras

OpenCV Scikit-Learn Raspberry PI

CUDA MLOps AWS Docker

Kubernetes Azure GCP Pandas

Matplotlib Flask REST Git SQL

Apache Spark Streamlit Redis

API Gateway React ONNX FastAPI

TOOLS & FRAMEWORKS

Vision: OpenCV · VGG16 · Grad-CAM · Mediapipe

Serving: ONNX · TorchServe · Flask · FastAPI

Training: CUDA · CuDNN · Early Stopping · W&B;

Data: Pandas · NumPy · Spark · Matplotlib

Deploy: Docker · Kubernetes · CI/CD · Linux

Cloud: AWS · Azure · GCP

IoT/Edge: Raspberry Pi · Jetson Nano · TPUs

KEY METRICS

94.7% Defect detection accuracy (VGG16)

50% API latency reduction via Redis

70% Manual work cut via CI/CD

MAE <10% AQI forecast precision

R² >0.85 Forecast model quality

5 cities Multi-language chatbot

EDUCATION

Bachelor in Computer Science NUST Islamabad

08/2022 – 08/2026 Islamabad, Pakistan